CryspLight full simulation: design

From the crystal to the recorded signal, for the comparison with the measurements

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Abstract

We are measuring the performance of BGO and CsI(Tl) crystals read out by an array of 64 SiPMs and ASIC front ends, at ambient temperature and at 196 K. This document specifies the simulation that accompanies the measurements. It starts from an existing optical Monte Carlo of the light transport in the crystal and carries it through to the signal that the acquisition records: the actual geometry of the SiPM array and the optical coupling, the wavelength of the light, the SiPM as a device with its gain, crosstalk and dark counts, and the electronics chain of the Citiroc and, later, of the Radioroc2. It defines the files exchanged between the stages of the simulation and with the measured data, the quantities on which simulation and measurements are compared, the tests that validate the code and the origin of every parameter.

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1 The problem and the simulation

We want to measure the performance of BGO and CsI(Tl) crystals as gamma-ray detectors at ambient temperature and at −77 °C (196 K). The main parameters are the number of photoelectrons per MeV —and from there the overall efficiency of the system to transform photons into photoelectrons—, the energy resolution, the scintillation lifetime, the uniformity of the response over the crystal and, eventually, the position resolution. The crystals are 48×48 mm^2 in cross section and two radiation lengths thick, polished, wrapped in Teflon and coupled through optical grease to an array of 64 SiPMs of 6×6 mm^2. The SiPMs are read out in two ways, one after the other on the same array: by two Citiroc 1A chips of 32 channels each, which store one charge per channel, and by a CAEN DT2745 digitizer of 64 channels, which records the waveform of every channel. In a second phase a Radioroc2 of 64 channels replaces the Citiroc. The cold measurements use the digitizer only, for a reason given in Section 7. A ^{22}Na source on the axis of the crystal, 30 mm from the face opposite the SiPMs, provides gammas of 511 and 1275 keV. Table 1 gives the three configurations. The ASICs stay at ambient temperature in all of them; in the cold configuration the crystal and the SiPMs are inside a cryostat and the signals come out through cables and feedthroughs.

	BGO ambient	CsI(Tl) ambient	BGO cold
crystal $x \times y \times z$ (mm)	$48 \times 48 \times 22.4$	$48 \times 48 \times 37.2$	$48 \times 48 \times 22.4$
thickness in X_0	2 ($X_0 = 11.2$ mm)	2 ($X_0 = 18.6$ mm)	2
temperature	laboratory, about 295 K	laboratory	196 K
light yield (ph/MeV)	7000 to 8200[a]	54 000	14 000 to 16 000[a]
decay time	300 ns, plus a minor 60 ns component	730 ns (64%) and 3340 ns (36%)	2.7 µs [b]
emission peak	480 nm	550 nm	480 nm
refractive index	2.15	1.79	2.15
intrinsic resolution, FWHM at 511 keV	5%[c]	5%	5%[c]
source	^{22}Na, on axis, 30 mm from the face opposite the SiPMs		
readout	Citiroc 1A ×2 and DT2745, in sequence	Citiroc 1A ×2 and DT2745, in sequence	DT2745

Table 1: The three configurations. [a] Not known well enough to be fixed: the ambient yield follows from the temperature dependence of the BGO light yield (Gironnet et al.) applied to the value at 195 K; the cold yield lies between the 14 000 ph/MeV of the calibration against the Ding et al. cube and the value implied by the longer decay time measured on our crystal. The ratio of photoelectrons between the warm and the cold runs will fix it. [b] Measured on this crystal (2.72 ± 0.01 µs at 196 K). [c] Less well known than for CsI(Tl); taken equal to it, and to be compared with the data once the statistical and noise terms are subtracted.

A realistic simulation is needed for two purposes. Before the data taking, it tells us how much light each channel will receive and therefore how to choose the operating parameters, such as the overvoltage of the SiPMs. After the data taking, the simulation provides a reference for the data. The simulation must therefore, model accurately the details of the actual detector.

The optical part of this simulation already exists: CryspLight, an optical Monte Carlo code in Julia and designed to run on a GPU, is able to transports the scintillation photons in a wrapped crystal read by an array of SiPMs. The initial implementation assumed an idealised readout, in which the SiPM array covers the whole face without gaps and the light is monochromatic and detected with a single efficiency.

The existing simulation stops at that idealised readout face. This document describes how it is carried through to the recorded signal. The ideal face is replaced by the actual array, so that a photon may fall between two SiPMs, and it reaches the active surface through the grease and the window of the device. Each photon is given a wavelength, because the efficiency with which a SiPM detects it depends on it. Each SiPM becomes a device that turns its photoelectrons into an electrical signal, with its gain, its crosstalk and its dark counts. And the electronics that measures that signal is simulated, so that the result is the value the acquisition would have recorded. How these extensions are added to the existing code is described in Section 12.

1.1 The measured data

With the Citiroc, the acquisition records for every event one charge per channel, in ADC counts: the value stored by the peak detector of the chip on its high-gain path (the chip holds a low-gain value as well, but our acquisition does not read it). The hit register of the chip, which says which channels fired their discriminator, is recorded as well. With the digitizer, the acquisition records for every event the waveform of each of the 64 channels: a fixed number of samples taken every 8 ns, a fixed part of them before the trigger, each sample a 16-bit count. The first step of the analysis of a digitizer run reduces each waveform to a baseline, estimated on the samples before the trigger, and a charge, the sum of the samples in a gate that starts at the trigger, with the baseline subtracted; it also notes which channels crossed the trigger threshold. For every run, with either readout, the acquisition records the conditions: how the SiPMs were biased (the bias voltage and the breakdown voltages measured in situ), how the electronics was configured (for the Citiroc its thresholds, hold delay, shaping time and gains; for the digitizer its gain, thresholds, record length and pre-trigger) and the temperatures. The first step of the analysis converts these files, charges or waveforms, to an HDF5 format described in Section 8, in which every event is one charge per channel whatever the readout; the simulation writes its results in the same format, so that the same analysis can be run on both.

1.2 The three stages of the simulation

The simulation runs in three stages, each one feeding the next through a file.

1. **Optical simulation.**
    - It generates the gammas from the source, follows them to their energy deposits in the crystal, emits the scintillation photons and transports them on the GPU until they are absorbed or reach a SiPM.
    - It writes one file per run with, for every event, the energy and position of each interaction of the gamma in the crystal, the number of scintillation photons emitted and the number detected, and one record for every detected photoelectron: the channel that detected it, the time of detection and the wavelength of the photon.
2. **SiPM and electronics simulation.**
    - It reads the photoelectron records and applies the response of the SiPM, which turns every photoelectron into a current pulse of a given shape, multiplied by the gain and accompanied by crosstalk and afterpulsing, and adds the dark counts of the device.

- It then applies the chain of the readout. For the Citiroc the pulses are amplified and shaped, compared with the discriminator thresholds, their maximum is stored by the peak detector until the hold, and the stored value is converted to ADC counts. For the digitizer the pulses are compared with the trigger threshold and integrated in the gate and the baseline window of the run, with the noise of the sampled signal, which gives directly the charge that the analysis of a measured waveform produces (Section 7.2).
- It writes one file per run with the same content as a measured run (for every event and channel, the charge in ADC counts, and the second path and the discriminator flag when the acquisition records them), in the format of Section 8, together with the energy deposits and the photoelectron counts of the first stage, which the analysis needs to interpret the result.

3. **Analysis.**

   - It reads a file in that format, simulated or measured, and extracts from it the quantities we compare: the gain of each channel from its single-photoelectron spectrum, its pedestal and noise, the energy spectrum of the sum of the channels and its resolution, and the maps of the response over the array.
   - It runs on simulated data for the time being and on the measured data once they exist. Whether it is written in Python or in Julia is left open: the loop over many events may require Julia.

Writing a file between the stages separates the slow part of the simulation from the fast one. The optical transport takes minutes on the GPU for a run of 10^5 events; the second and third stages take seconds. A change in the parameters of the SiPMs or of the electronics, for instance when they are adjusted to reproduce the data, therefore requires rerunning only the last two stages, and the photoelectron records are reused.

2 Geometry

2.1 Crystal and wrapping

The crystal is a rectangular box, 48×48 mm^2 in cross section and 22.4 mm (BGO) or 37.2 mm (CsI(Tl)) thick, polished on its six faces. Five faces are wrapped by hand in several layers of PTFE tape. The tape does not touch the crystal everywhere, so the model places a thin layer of air between the two. A photon that reaches a wrapped face is first reflected or transmitted at the crystal-to-air interface according to Fresnel's equations; only the transmitted fraction reaches the Teflon, which returns it diffusely with a reflectivity of 0.99, the value measured by Janecek and Moses for three or more layers of tape. The polish is described with the UNIFIED model of Geant4: the normal to the surface seen by each photon deviates from the geometric normal by a Gaussian angle of 1.3° rms, a value obtained for crystals of the same finish when the optical simulation was calibrated.

The air gap has a consequence that dominates the light collection. A photon whose direction is beyond the critical angle (27.7° in BGO, 34° in CsI(Tl)) on the face it meets is totally reflected, and since the faces of the box are parallel, a photon beyond the critical angle on all three pairs of faces stays trapped until the roughness of the surface, Rayleigh scattering in the bulk or absorption changes its direction or ends its life. The roughness is therefore not a detail: it sets how fast the trapped light leaks towards the readout face.

2.2 Coupling and window

The sixth face is coupled through optical grease ($n = 1.42$) to the SiPM tiles, whose photosensitive surface is covered by 0.15 mm of silicone resin ($n = 1.57$, from the datasheet). A photon leaving the crystal through this face crosses two interfaces, crystal to grease and grease to resin, and at each one it is reflected or transmitted according to Fresnel's equations. The first interface dominates (from 2.15 or 1.79 down to 1.42); the second reflects 0.3% at normal incidence. A reflected photon returns to the crystal and continues its life. The step from the resin into the silicon is not needed, since the detection efficiency quoted by Hamamatsu includes it. The detection efficiency is applied once the photon is inside the resin above an active area, as a function of its wavelength and of the overvoltage (Section 6), and independently of its angle of incidence. This independence is an assumption: the datasheet measures the efficiency at normal incidence from air, whereas our photons arrive from the grease at all angles up to the critical one. It is listed in Section 13.

2.3 The SiPM array

The array is made of four Hamamatsu S14161-6050HS-04 tiles, each with 4×4 channels of 6.0 mm active width on a 6.2 mm pitch. The tiles are butted, and since the active area stops 0.2 mm short of the package edge, the active areas of two channels across a tile boundary are 0.4 mm apart. With the origin at the centre of the matrix, the 64 centres are at ± 3.2, ± 9.4, ± 15.6 and ± 21.8 mm in each coordinate.

This geometry differs from an ideal array in three ways. The matrix is 50.0 mm wide, so the outer channels overhang the crystal by 1.0 mm and only 5.2 mm of their width faces it. Inside the 48×48 mm^2 footprint the active area is 93.4% of the total, the rest being the 0.2 mm gaps between channels, the 0.4 mm seam that crosses the centre in both directions and the overhang. And a photon transmitted into the resin above one of these gaps meets the surface of the package between the channels, whose reflectivity we do not know; the model takes it diffuse with a reflectivity of 0.5 and Section 9 says how the data will constrain it.

The array enters the simulation as a list of active rectangles, each with its channel number, so that the same code describes this array, an ideal one or any future layout. The channel numbering follows the cabling of the acquisition and is part of the description of each run (Section 8).

3 Source and gamma transport

The ^{22}Na source is a point on the axis of the crystal, 30 mm from the centre of the face opposite the SiPMs. In each decay it emits a gamma of 1274.5 keV and, in 90.3% of the cases, a positron that annihilates into two back-to-back gammas of 511 keV. The directions are isotropic. The simulation generates whole decays, draws the directions of the 1275 keV gamma and of the 511 keV pair, and transports the gammas whose line meets the front face. The face covers 13% of the sphere seen from the source, so the 1275 keV gamma enters in 13% of the decays; the 511 keV pair enters in 25% of the decays with a positron, twice as often, because either partner can be the one aimed at the face, and never both. Keeping the gammas of one decay together matters for the spectrum: when the 1275 keV gamma and a 511 keV gamma both deposit energy in the same event, their energies add, which produces the sum peak at 1786 keV, and the Compton continuum of the 1275 keV line lies under the 511 keV photopeak with its real weight. The gammas arrive at angles of up to 49° at the corners of the face, so the depth of the interactions differs from that of a beam at normal incidence.

Inside the crystal the gammas are followed by the transport code of PTCrysp, which simulates photoelectric absorption and Compton scattering with cross sections tabulated up to 10 MeV. Pair production, open above 1.022 MeV, is treated as a local absorption; at 1275 keV it is below

1% of the total cross section. Three effects are not simulated: the fluorescence that follows a photoelectric absorption on the K shell of bismuth, whose 77 keV X-ray can escape when the interaction is within a fraction of a millimetre of a surface and produces an escape peak at 434 keV; the Doppler broadening of the Compton scattering; and the scattering of the gammas in the source holder and in the walls of the cryostat. They are listed in Section 13.

4 Scintillation

Each energy deposit emits a number of photons drawn from a Poisson distribution whose mean is the light yield of the material times the energy deposited (Table 1). The crystal itself broadens the photopeak beyond the photon statistics. The light yield per unit energy of a scintillator depends on the energy of the electron that deposits it, and a gamma of a given energy is converted into electrons of different energies depending on how it interacts (one photoelectron, or a chain of Compton electrons followed by a photoelectron), so the total light fluctuates from event to event even for the same deposited energy. This intrinsic resolution is known for CsI(Tl), close to 5% FWHM at 511 keV, and less well for BGO. The non-proportionality curve of BGO has been measured (Gentile et al., 2015, and the references therein): relative to 662 keV, the light per keV is 73% at 23 keV, 89% at 60 keV, about 92% between 80 and 120 keV, 93% at 186 keV and 98.6% at 356 keV. A 511 keV deposit that proceeds through Compton electrons of 100 to 300 keV therefore yields a few percent less light than a single photoelectron, which is the origin of an intrinsic term of the same order as in CsI(Tl). The simulation takes 5% FWHM for both crystals, as a parameter, and applies it as a Gaussian smearing of the light of each event; the direct measurements that separate this term from the photon statistics (Moszynski et al.) remain to be checked. A more faithful treatment is possible, because the gamma transport records every deposit with its energy: weighting the light of each deposit by the curve at the energy of its electron produces the Compton-chain part of the intrinsic resolution for every line and for the continuum, leaving only the fluctuations within a single electron track to a smaller smearing. It is planned as a refinement once the first comparison with the measured widths of the 511 and 1275 keV lines says whether it is needed.

Each photon is emitted at a time after the deposit drawn from the decay profile of the material: a single exponential for BGO, and for CsI(Tl) the sum of a 730 ns component carrying 64% of the light and a 3340 ns component carrying the rest. The two components of CsI(Tl) matter for the electronics, whose shaper is much shorter than either, because the stored signal follows the rate of arrival of the photons at the beginning of the pulse, which the fast component sets. For BGO at 196 K the decay time is the 2.7 µs measured on this crystal, in a material description of its own, so that the value measured on the Ding cube stays with the calibration it belongs to.

Each photon is also given a wavelength drawn from the tabulated emission spectrum of the material. The refractive index of the crystal, the Rayleigh scattering length (which scales as λ^{-4} from the calibrated value at the emission peak) and the detection efficiency of the SiPM are then evaluated at that wavelength. The bulk absorption length keeps its calibrated scalar value.

5 Optical transport

The photons are transported one by one on the GPU, each with its own stream of random numbers, so that a run is reproducible photon by photon whatever the number of threads. A photon moves in straight lines inside the box; at each step the code decides whether it is absorbed in the bulk or scattered by Rayleigh scattering before reaching a face, and at the face it applies the surface model described in Section 2: Fresnel reflection or transmission with the roughened normal, the air gap and the Teflon on the wrapped faces, the grease and the resin on

the readout face. The photon ends its life when it is absorbed in the bulk, in the wrapping, in the package between the SiPMs or in a SiPM without producing a photoelectron, or when it is detected. The code checks that every photon ends in exactly one of these ways.

Three extensions of the transport are needed for the actual detector, and all three have been described above: the array of active rectangles with the reflecting surface between them, the second interface and the resin layer (the lateral displacement of the photon inside 0.15 mm of resin is negligible against the 6 mm pitch and is not tracked), and the wavelength of each photon with the tabulated refractive index, Rayleigh length and detection efficiency.

The optical simulation writes one file per run. For every event it records the energy and position of each interaction of the gammas in the crystal, with the number of interactions and the type of the first one, the number of scintillation photons emitted and the number detected, and the detected photoelectrons as a list of records of channel, detection time and wavelength, stored consecutively for all events with an index giving where each event starts. A record occupies 7 bytes, so a 511 keV event takes about 70 kB in CsI(Tl) and 15 kB in BGO, and a run of 10^5 events per configuration takes a few gigabytes of disk and a few minutes on the GPU.

6 Response of the SiPM

The SiPM is the S14160-6050HS element of Hamamatsu (datasheet KAPD1064E, November 2024); the S14161 tiles carry the same element. The model describes each channel separately.

Breakdown voltage and overvoltage. The breakdown voltage is 38.0 V at 25 °C and rises by 34 mV per kelvin, so it is near 34.5 V at 196 K; a measurement on one of our channels at that temperature gave 34.3 V. Within a tile the channels differ by 0.1 V typically and 0.2 V at most. The simulation takes the breakdown voltage of each channel from the run when it has been measured, as the operating recipe prescribes, and otherwise draws it from that spread. The overvoltage of a channel is its bias minus its breakdown voltage, after the per-channel trim of the ASIC input DAC if it is used.

Gain. The charge of one avalanche is proportional to the overvoltage: 0.93×10^6 electrons per volt, which gives 2.5×10^6 at 2.7 V.

Detection efficiency. The probability that a photon arriving on the active area produces an avalanche is the product of a spectral factor, read from the datasheet curve and normalised to one at its maximum of 450 nm, and the efficiency at 450 nm as a function of the overvoltage: 37% at 1.7 V, 50% at 2.7 V, 54% at 3.2 V and constant above, because the probability of triggering an avalanche saturates.

Crosstalk and afterpulsing. An avalanche emits a few photons that can fire a neighbouring cell at the same time. The probability grows linearly with the overvoltage, from 2.5% at 1.6 V to 13% at 3.7 V; the model applies it as a geometric number of extra simultaneous avalanches per primary one. Afterpulsing, the delayed release of a carrier trapped during the avalanche, is only qualified as low in the datasheet; measurements on devices of the same technology give about 1% at 3 V, growing with the square of the overvoltage, with delays of tens to hundreds of nanoseconds. The model uses 1% at 3 V with a time constant of 100 ns, as an adjustable parameter, and reduces the charge of the afterpulse by the fraction of the recharge of the cell completed at its delay, with the recovery time given with the pulse shape below. The parameter cannot be measured with a charge-only readout and its effect, a correlated per-cent addition to every channel absorbed by the single-photoelectron calibration, is below what the comparison can resolve.

Dark counts. Thermal carriers start avalanches at a rate that grows with the overvoltage like the detection efficiency and with the temperature by a factor 2 every 9 K. The datasheet dark current at 25 °C and 2.7 V, 2.5 μA, corresponds to 6×10^6 avalanches per second per device once the crosstalk is taken out; at 196 K and 6.5 V we measured 8 to 15 kHz per channel. The model adds dark avalanches, with their crosstalk, over the whole acquisition window of every event.

Pulse shape and cell occupancy. One avalanche produces a current pulse whose shape depends on the input impedance of the electronics. The pulse measured on our channels into 100 Ω rises in 16 ns, is 24 ns wide at half maximum and decays in 50 ns; it is the default shape, scaled by the gain, and it is replaced by a measurement at the Citiroc input when one exists. A device has 14 331 cells, and a cell that has just fired needs about 50 ns to recover; the model keeps track of this, although only the brightest CsI(Tl) channel comes within a percent of being affected.

The output of this stage, for every channel, is a train of avalanches with their times and charges, which is the input of the electronics.

7 Electronics

7.1 The Citiroc 1A

Two chips read the 64 channels, 32 each, with the channels assigned by the cabling of the run. The chain of each channel, as the datasheet (DS7004, version 2.5) describes it and as the simulation reproduces it, is the following.

1. **Input.** The train of avalanches becomes a voltage across the termination of the input, which the datasheet specifies as 50 Ω in series with 100 nF. The input DAC of the channel does not touch the signal; it shifts the bias of the SiPM and therefore enters the model through the overvoltage.

2. **Preamplifiers.** Two voltage preamplifiers work in parallel on each channel: a high-gain one, with 15 pF of input capacitance and a gain adjustable between 10 and 600, and a low-gain one, with 1.5 pF and a gain between 1 and 60. The simulation gives each channel its own gain, drawn around the nominal value with an adjustable spread.

3. **Shapers.** Each preamplifier feeds a CR-RC^2 shaper with a peaking time between 12.5 and 87.5 ns, set in steps of 12.5 ns for the whole chip. A CR-RC^2 shaper is one differentiating stage followed by two integrating ones, all with the same time constant θ. The preamplifier integrates the current of the SiPM, so the output of the shaper for one avalanche is the response of the chain to a step, the semi-Gaussian $(t/\theta)^2 e^{-t/\theta}$, whose peak is at $t_p = 2\theta$; θ runs from 6.25 to 43.75 ns over the available settings. The shaped signal of an event is the convolution of the current of the avalanche train with that response. The datasheet states the order of the shaper and nothing more; the exact response is a parameter to be checked against a single-photoelectron waveform if one can be probed on the test board.

4. **Noise.** A Gaussian noise series is added after the shaper, with the spectrum measured on the pedestal or control events of the run. Until a run exists, the noise is white with an rms of 0.1 photoelectron on the high-gain path, which is the signal-to-noise ratio of 10 for one photoelectron quoted by the datasheet.

5. **Trigger.** A fast shaper with 15 ns peaking time, on the high-gain path, feeds two discriminators whose thresholds are common to the chip (10 bits) with a 4-bit trim per channel. One of them provides the trigger output of the channel; the other fills the hit register and

the OR of the 32 channels. The simulation records, for each channel, the first time the fast-shaped signal crosses the threshold; the trigger of the acquisition is then a configurable function of these times (the OR of one chip, the OR of both, or a majority of N channels within a coincidence window), as the FPGA builds it in the real system.

6. **Peak detector and hold.** When the chip triggers, each channel starts storing the maximum of its shaped signal, on the high-gain and the low-gain path separately, and keeps it until the hold arrives, a fixed delay after the trigger (up to 500 ns in our acquisition). Each of the 64 channels stores its own maximum, and a channel that receives little light stores the maximum of its noise within the hold window; this gives a positive bias to the dim channels which the simulation reproduces because it stores the maxima the same way. The alternative mode of the chip, in which the signal is sampled at a fixed time instead of at its maximum, is available in the simulation as well.

7. **Readout and conversion.** The stored values are read out through a multiplexer at up to 5 MHz and digitised by an external ADC. The simulation applies the gain and offset of the ADC, its quantisation (12 bits by default) and the saturation of the chain, which is reached either at 400 pC of input charge or at the full scale of the ADC, whichever comes first on each path. The dead time of the readout is recorded but does not reduce the number of events, because the rates with our source are low.

The stored value of a channel is the peak of the shaped signal, calibrated in photoelectrons by the peak that one avalanche produces. For a scintillation pulse much longer than the peaking time the shaped signal does not reach the amplitude that all the photoelectrons would give if they arrived together: it is the convolution of the arrival rate with the pulse of one avalanche and with the shaper response, and its maximum, divided by the maximum of the same convolution for a single avalanche, is the fraction of the photoelectrons that the measurement keeps. With the peaking time at 87.5 ns and the avalanche pulse of Section 6 (16 ns rise, 50 ns fall), the fraction is 0.39 for BGO at ambient temperature, 0.15 for CsI(Tl), whose 730 ns component carries 64% of the light, and 0.07 for BGO at 196 K with its 2.7 µs decay. For a decay time τ much longer than the peaking time the fraction is the ratio of an effective window to τ; the window is $1.85\,t_p = 160$ ns for an infinitely short avalanche pulse and grows with the width of that pulse, because a wider pulse lowers the peak of the single avalanche that serves as the unit while the slow signal, which the shaper sees as a rate, keeps its amplitude: it is 185 ns for the 50 ns fall and 230 ns for a 100 ns fall. The pulse at the Citiroc input is therefore the parameter that fixes these fractions, which is why it has to be measured. The rise of the scintillation itself, below a nanosecond in BGO and some tens of nanoseconds in CsI(Tl), is short compared with the window and changes nothing. The photon statistics of the energy measurement is that of this fraction and not that of all the photoelectrons; the simulation computes it with the 64 channels, their noise and their dark pulses.

7.2 The DT2745 digitizer

The CAEN DT2745 samples its 64 channels every 8 ns with 16 bits over a full scale of 4 V, so one count is 61 µV, through an analog bandwidth of 20 MHz and a gain programmable between 1 and 100 for each group of 16 channels. Each SiPM is connected directly to the 100 Ω input of one channel; there is no preamplifier and no shaper. The pulse of one avalanche, measured on our channels, rises in 16 ns, is 24 ns wide at half maximum and decays in 50 ns, and the bandwidth of the digitizer is what sets that shape. Its peak is 3.2 counts per volt of overvoltage at gain 1, and its area, the quantity the analysis measures, is 4.4 times its peak in count-samples. Each channel fires when its sampled signal rises above its baseline by a threshold in counts, and the acquisition takes an event when N channels fire within a coincidence window; it then stores, for each of the 64 channels, a record of fixed length with a fixed number of samples before

the trigger. The analysis of a record estimates the baseline of the channel as the mean of its pre-trigger samples and defines its charge as the sum of the samples in a gate that starts at the trigger and lasts long enough to contain the scintillation decay, minus the baseline times the number of samples in the gate. The single-photoelectron area, measured on dark pulses, converts that charge into photoelectrons.

The simulation of this readout does not build the waveforms. It computes, from the train of avalanches of each channel, the charge that the analysis would have extracted from the waveform, and writes it in the same event file as the Citiroc charges. This is possible because every operation between the avalanches and the charge is linear except the trigger. The sampled waveform of a channel is the sum of the pulses of its avalanches, each pulse scaled by the charge of its avalanche, plus the noise of the digitizer. The charge is a weighted sum of samples: weight one in the gate, weight minus the ratio of the gate length to the baseline length in the baseline window, zero elsewhere. Applying that weighted sum to the pulse of one avalanche gives a number that depends only on where the avalanche falls: its full area if it falls inside the gate, a fraction of it if it straddles an edge, minus the scaled area if it falls in the baseline window. The charge of the event is therefore the sum over the avalanches of the channel, scintillation and dark alike, of their charge times this number, plus the weighted sum of the noise samples. For white noise with an rms σ_s per sample the latter is a Gaussian with rms $\sigma_s\sqrt{N_g + N_g^2/N_b}$, with N_g and N_b the numbers of samples in the gate and in the baseline window, and one draw per channel and event is all the noise the charge needs. In units of photoelectrons this noise is 0.6 for a gate of 1 µs and 2 for a gate of 10 µs at 5 V of overvoltage and gain 1; the low-frequency noise and the correlation between the channels of one group seen in our earlier runs are not in this model, and they are added, as a second Gaussian common to the 16 channels of a group, when the new runs have measured them.

The trigger is the one non-linear step and the only one that needs the shape of the pulse. The simulation evaluates the sampled amplitude of each channel, the sum of the pulses of its avalanches on the 8 ns grid, around the times where avalanches occur, and records the first sample at which it exceeds the threshold. The trigger time of the event is then the time at which N of these crossings fall within the coincidence window, as in the acquisition, and it is the origin from which the gate and the baseline window are placed. The threshold decides which events are taken and, at ambient temperature, how often the dark pulses take one instead. The single dark photoelectron is not the problem: a dark avalanche fires neighbouring cells with the crosstalk probability, 19% at 5 V, so one dark pulse in 700 carries four photoelectrons or more, and over the 64 channels such pulses come about once per microsecond; a threshold of two or three photoelectrons triggers on them in every window. The emulation gives the efficiency for the photopeak and for the Compton events and the rate of dark triggers as a function of the threshold and of the majority. For ambient BGO at 43 V and gain 1 the brightest channel of a 511 keV event peaks at 12 to 15 photoelectrons on the sampled waveform, and a threshold of 7.5 photoelectrons (120 counts) with a majority of two keeps every photopeak event and 99% of the events between 200 and 300 keV with dark triggers below 1 kHz; the same holds for the Citiroc at 3 V with a single-channel threshold of 8 photoelectron-equivalents on its fast path.

What goes into the event file is then the same as for the Citiroc: one charge per channel, as a 32-bit integer in count-samples, the set of channels that crossed the threshold in place of the hit register, and the trigger time. The single-photoelectron area and the noise per channel go into the calibration group. The saturation of the input, at a peak amplitude of 4 V divided by the gain, is checked on the brightest channel and flagged; it is not reached in our configurations. A subsample of events, a thousand per run, is also written as full waveforms, built with the same pulses and noise, for figures and for a visual comparison with the measured records; they are not used by the analysis.

The gate is a choice of the analysis that the simulation informs, in its position as in its length. The emulation scanned both: opening the gate earlier than 50 ns before the trigger

brings nothing, because the trigger fires within a few tens of nanoseconds of the first light, and every additional nanosecond of gate collects dark pulses; at 5 V of overvoltage the best gate is 1 µs for ambient BGO (9.3% FWHM, 10.2% at 2 µs) and 3 µs for CsI(Tl) (5.9%, 6.2% at 4.5 µs). At that overvoltage the dark pulses in the gate, 34 avalanches rms over the 64 channels for BGO, and the excess noise of 19% crosstalk make the digitizer the least precise of the three front ends on ambient BGO; at 3 V both terms shrink, which is why the operating point with the digitizer gain at 4 and 3 V is the preferred one. The bench has two operating points: gain 4 of the digitizer with 3 V of overvoltage, if the noise of the digitizer does not grow with its gain, which gives one photoelectron of 38 counts and the crosstalk and dark rate of 3 V; or gain 1 with 5 V, 16 counts per photoelectron, which the first runs use when the noise check is not available. The length of the gate is the other choice. The gate must contain the scintillation decay, but at ambient temperature every channel collects dark photoelectrons at 6 to 8 MHz, and in a gate of 10 µs the 64 channels gather about 5000 of them against 1650 of signal in BGO: their fluctuation alone is 4% rms. A gate of 1 µs for ambient BGO, whose decay is 300 ns, and of 4 to 5 µs for CsI(Tl) brings that term below 1%. In the cold the dark rate is a thousand times lower and the gate can take the whole 2.7 µs decay with no penalty, which is why the digitizer, and not the Citiroc, reads the cold runs: the Citiroc stores 0.07 of the photoelectrons of cold BGO (Section 7.1); the digitizer, with a gate of 10 µs, stores them all.

7.3 The Radioroc2

A single chip reads the 64 channels (datasheet version 2.0, January 2025). Its chain has the structure of the Citiroc one, with the differences listed below, and the simulation applies the same code with these differences as settings.

1. **Preamplifiers and shapers.** Two charge preamplifiers per channel, with gains between 5 and 80 on the high-gain path and 0.5 and 8 on the low-gain one, each followed by a shaper whose peaking time is set channel by channel: 20 to 300 ns in steps of 20 ns, or, in the slow mode, 120 ns to 1.8 µs in steps of 120 ns. The order of the shaper is not stated; the simulation takes CR-RC^2, as for the Citiroc, until a single-avalanche waveform is measured. The slow mode is the natural choice for CsI(Tl) and for cold BGO, whose 2.7 µs decay through 1.8 µs of shaping stores 0.55 of the photoelectrons (Section 7.1); at ambient temperature a long shaping also collects more dark pulses, about 14 photoelectrons per channel in the effective window of the longest setting, so that the shaping time has an optimum per crystal which the emulation gives.

2. **Noise.** The datasheet gives a signal-to-noise ratio above 10 for one photoelectron on the high gain, with no shaping time attached. With 2 nF of device capacitance at the input the series noise dominates, and it falls as the inverse square root of the peaking time, while the dark pulses, which the model generates explicitly, grow with it. The simulation takes 0.1 photoelectron rms at 100 ns scaling as $(100\ \text{ns}/t_p)^{1/2}$, a parameter to be replaced by the pedestals of the runs.

3. **Triggers.** Each channel has a time preamplifier of 200 MHz bandwidth with two discriminators, T1 and T2, of independent thresholds common to the chip with a per-channel trim, and a charge discriminator TQ on the output of the low-gain shaper. The time preamplifier sees the avalanche almost unshaped, so T1 and T2 respond to the instantaneous amplitude within a few tens of nanoseconds: for ambient BGO the brightest channel of a 511 keV event reaches only 4 to 5 photoelectrons there, while the dark pulses with crosstalk reach the same once per microsecond over the array, and a single time threshold cannot separate them. The charge trigger sees the integrated light of the event, hundreds of photoelectron-equivalents for the photopeak against a handful for a dark pulse, and is the trigger the simulation takes by default for the slow scintillators; T1 and T2, single

channel or in a majority built by the FPGA of the board, are emulated as the comparison. The pulse at the time preamplifier is not the one measured through the 20 MHz digitizer; the simulation takes a 5 ns rise and a 20 ns fall as parameters, which only enter the T1 and T2 numbers.

4. **Acquisition trigger, hold and peak detectors.** The peak detectors of all channels are opened by one acquisition trigger, selectable among the chip-wide OR of T1, of T2 or of TQ, the local trigger of each channel, or an external signal; the hold follows it after a delay set on the chip between 2 ns and 2.6 µs, or comes from outside. Each path stores the maximum of its shaped signal from the trigger to the hold, as in the Citiroc, with the same positive bias of the dim channels.

5. **Readout and conversion.** The stored values of the two paths leave through two analog multiplexers at 200 ns per channel, 12.8 µs for the 64 channels, to the ADC of the acquisition board, which the simulation takes as the 12-bit converter of the evaluation board. The low-gain path covers 2000 photoelectrons at a gain of 10^6, about 320 pC, at its default preamplifier gain of 0.53; the high-gain path, whose default gain is 5.3, saturates ten times lower. The ceilings scale inversely with the preamplifier gains, but the high gain cannot be set below 5, so its ceiling cannot exceed about 76 avalanches at our SiPM gain of 2.8×10^6, which the brightest channel of a 511 keV event in ambient BGO exceeds in a quarter to a half of the events; the chip is therefore operated as designed, both paths read and combined, the high gain for every channel below its ceiling and the low gain above it, and the simulation stores that dual-gain value. Each channel can instead route its shaped signal to a pin of its own, for a parallel digitisation that the present set-up does not use.

8 Files

8.1 The photoelectron file

The optical simulation writes its photoelectron records in one HDF5 file per run. The records of all events are stored consecutively in three arrays, `channel` (one byte), `t_ns` (a 32-bit float) and `lambda_nm` (16 bits), and an array `offset` of length one more than the number of events gives the position of the first record of each event. The event index is the same as in the file of interactions and photon counts written by the same run. The attributes of the file record the configuration of the run and the version of the code.

8.2 The event file, common to measurements and simulation

The measured runs are converted, as the first step of the analysis, into one HDF5 file per run with the structure of Table 2; the electronics simulation writes the same structure directly, with the additional group `/mc` of the simulated quantities. Arrays are stored event-major, so that a reader obtains an array of n events by 64 channels without transposition.

9 Observables for the comparison

The analysis computes the following quantities on a simulated and on a measured file alike, and each of them tests a different part of the model.

1. **Single-photoelectron spectra.** Built from dark pulses, channel by channel, they give the gain of each channel and its spread; repeated at several bias values, they give the breakdown voltage of each channel, and through it the temperature of the SiPMs. They test the gain model.

group	content
`/run`	Attributes describing the run: date, crystal and material, temperature readings and where they were taken, bias per chip, breakdown voltage measured per channel, overvoltage per channel, discriminator thresholds, hold delay, shaping times, preamplifier gains, ADC, trigger definition; for a digitizer run its gain, record length, pre-trigger, threshold and majority, and the gate and baseline window used to define the charge; and the version of the converter or of the simulation. For a simulated run, the full configuration of the simulation with the origin of every parameter.
`/channels`	One entry per channel: the position of its centre in millimetres, its tile, its chip and channel number on the chip, and its status (live, dead or noisy). This group ties the acquisition channel to the geometry of Section 2.3.
`/calib`	The single-photoelectron gain of each channel, in ADC counts per photoelectron with its error, and the mean and rms of its pedestal, on each path that is recorded; its dark rate. Measured on the run or on its calibration runs; for a simulated run, the values used.
`/events`	For every event: the charge of the 64 channels, as 16-bit integers of ADC counts for the Citiroc and 32-bit integers of count-samples for the digitizer; the hit register of the 64 channels, which for the digitizer is the set of channels that crossed the threshold; the trigger time; and flags for saturation and for pile-up. A simulated run may also carry the low-gain charge, which the Citiroc holds but our acquisition does not read, and which the Radioroc2 acquisition will.
`/mc`	Simulated runs only: the quantities written by the optical simulation for the same events (positions and energies of the interactions, photons emitted and detected, the integrated photoelectron map), so that any observable can be examined against them.

Table 2: Structure of the event file.

2. **Pedestals and control charges.** They measure the noise of each channel and of the sum. They are an input of the simulation, not a prediction of it.

3. **Dark rate per channel**, against temperature and bias.

4. **The energy spectrum** of the equalised sum of the 64 channels: the 511 and 1275 keV lines, the sum peak, the Compton continua and the backscatter region. The ratio of the two photopeak positions, 2.5 if the chain is linear, tests the dynamic range and the trigger. The width of the 511 keV line, decomposed into the noise measured from the pedestals, the statistics of the stored fraction and the remainder, tests the electronics model and measures the intrinsic resolution.

5. **The light yield.** The photoelectrons at 511 keV, converted into photons per MeV through the simulated collection efficiency and detection efficiency. The ratio between the warm and the cold BGO runs tests the temperature dependence of the yield.

6. **The brightest channel.** The distribution of the largest charge in an event, and its ratio to the total, test the light distribution near the readout face; in CsI(Tl) its tail shows the saturation of the chain.

7. **Channel maps.** The fraction of events in which each channel is hit and the mean charge it collects, compared with the simulated maps, test the response of the outer channels (which see only 5.2 mm of crystal), of the central seam and of the tile boundaries. The reflectivity of the surface between the channels and the roughness of the crystal are the two parameters these maps constrain.

8. **The seed-channel map.** The position of the 511 keV photopeak for the events grouped by their brightest channel, in units of the median over the channels. In a healthy array this is a map of the non-uniformity of the light collection, which the simulation must reproduce.

9. **Dependence on the overvoltage**, when a bias scan exists: light, crosstalk, resolution and saturated fraction against the overvoltage, to be compared with the curves of Section 6.

10. **The readouts on the same crystal.** The photoelectrons at 511 keV measured with the digitizer, which integrates the whole decay, and with the Citiroc or the Radioroc2, which store a fraction of them, must be in the ratio computed in Section 7.1 from the pulse shape and the shaping time: 0.39 for BGO and 0.15 for CsI(Tl) with the Citiroc, 0.69 for BGO with the Radioroc2 at 1.8 µs, at ambient temperature. This is a direct test of the model of the shaping chains with no free parameter.

11. **The two paths of the Radioroc2.** The high-gain path gives the single-photoelectron peak of every channel and the low-gain path the energy; on the channels that both paths see between 10 and 70 avalanches, the ratio of the two readings, channel by channel, is the low-gain calibration constant, nominally a tenth of the high-gain one and measured to better than a per cent in minutes of source data. The 1275 to 511 ratio of the low-gain sum then tests the linearity of that path alone.

12. **Trigger efficiency and gate length**, for the digitizer: the photopeak content and the trigger rate against the threshold, and the resolution against the length of the gate, compared with the simulated curves that fixed the settings.

10 Validation

The code is tested at each of its extensions, with tests that have an analytic answer or reduce a new feature to an old one.

- The array described as active rectangles reproduces, with the rectangles of an ideal array, the result of the ideal array; with the gaps and no reflection on the surface between the channels, the fraction of photons lost under uniform illumination equals the geometric dead fraction.

- When the grease and the resin have the same index the second interface reflects nothing; when the resin has the index of the crystal and there is no grease, the result is that of a crystal in optical contact with the sensor.

- A spectrum reduced to a single wavelength reproduces the monochromatic simulation.

- The photoelectron records, binned in time, reproduce the time-binned maps written by the run.

- The rate of gammas on the front face equals the solid-angle fraction, and the distribution of their incidence angles the analytic one.

- A single avalanche through the shaper gives the analytic CR-RC^2 peak at t_p; a Poisson train with a long decay time gives the stored fraction of Section 7 within its statistics; a peak detector fed with noise only gives the expected bias of the maximum of a Gaussian series; the saturation clips at the configured charge.

- The energy resolution of a simulated configuration agrees with the quadrature of its stated terms.

11 Parameters and their provenance

12 Implementation

The optical simulation is CryspLight, a Julia package whose transport kernel runs on the GPU of Apple computers through Metal and, identically, on the CPU. Its reference version of the transport and its GPU version are kept equal draw by draw, and a test proves that they produce the same result bit for bit on the CPU; every extension described here is written in both.

The state of the code before this work is kept under the git tag `pedsim`, and the extensions grow on the branch `fullsim`. Nothing in the existing behaviour is changed: each extension is switched on by a block in the configuration file of the run and switched off by its absence, in which case the code follows the old path with the same random numbers. A test runs a small configuration of the old kind after every change and compares its output with stored checksums. The existing configurations, output files and analysis scripts are left as they are; the new simulation has its own configurations under `fullsim_runs/`, grouped by measurement campaign, its own analysis under `fullsim_analysis/` and its documents under `docs_fullsim/`. The analysis is written in Julia, a library of readers, fits, spectra and maps and one script per study; the scripts compute and write files, HDF5 for arrays and CSV for tables, under `fullsim_analysis/results/`, and Python scripts draw the figures from those files, so that a figure is remade without recomputing and every number cited in a document is in a file produced by a tracked script. The SiPM and electronics simulation is a new module, `src/frontend/`. Simulation outputs go to the directory `output/`, which is not under version control, and the measured files stay outside the repository and are referred to by path from the configuration of each campaign.

The optical stage records every photon that reaches the resin above an active area, with the detection efficiency set to one in the transport, and the SiPM stage applies the efficiency by keeping each record with the probability of Section 6 for its wavelength and the overvoltage of the run. Thinning a Poisson process this way is exactly equivalent to applying the efficiency photon by photon during the transport, and it lets one optical run serve every overvoltage of a scan. The photoelectron file is written in chunks of decays as the run proceeds, so that a run of any length streams to disk; the random streams of the decays and of their photons are disjoint between chunks, and the gamma-level content of a run does not depend on how it was chunked.

The GPU imposes a few constraints on the extensions: the tables of the spectra and of the wavelength-dependent quantities travel in one buffer of 32-bit floats, the switches and scalars in one structure, and no new random number is drawn unless the corresponding extension is on.

13 Open items

- Whether the detection efficiency depends on the angle of incidence when the photons arrive from the grease rather than from air.
- The reflectivity and the specularity of the surface of the package between the channels.
- The light yield of this BGO crystal at ambient temperature and at 196 K, and whether its 2.7 µs decay time means that it was colder than the thermometer indicated.
- The intrinsic resolution of BGO.
- For the Citiroc: the exact response of the shaper, the noise spectrum, the droop of the peak detector during the readout, and the pulse shape at the input of the preamplifier with a device of 2 nF.

- For the Radioroc2: the order of the shaper, the noise, the behaviour of the 100 Ω input mode with 2 nF, and the acquisition board and ADC to be used; and whether its longest shaping, which stores 0.55 of the photoelectrons of cold BGO, is enough for the cold runs.
- For the digitizer: the noise below 1 MHz and its correlation within a group of 16 channels, and whether the gain stage lowers the noise per photoelectron, which decides whether the cold runs can be taken at 3 V of overvoltage with gain 4 rather than at 5 V with gain 1.
- In the gamma transport: fluorescence and escape of the K-shell X-rays, Doppler broadening, and scattering in the source holder and in the cryostat.
- The temperature of the SiPMs relative to the crystal in the cold run, which the breakdown voltage measured in situ resolves in practice.

parameter	value	source	status
crystal sizes	Table 1	setup	fixed
polish σ_α	1.3°	calibration of the optical simulation	fixed
Teflon	Lambertian, $R = 0.99$, air gap	Janecek & Moses	fixed
grease index	1.42	setup	fixed
resin index, thickness	1.57, 0.15 mm	datasheet	fixed
SiPM array	centres at $\pm 3.2, \pm 9.4, \pm 15.6, \pm 21.8$ mm	setup	fixed
reflectivity between channels	0.5, diffuse	assumption	to fit on maps
BGO yield ambient	7000 to 8200 ph/MeV	Gironnet curve, Moszynski	to fit on data
BGO yield 196 K	14 000 to 16 000 ph/MeV	Ding calibration; our decay time	to fit on data
BGO decay 196 K	2.7 µs	our measurement	fixed
BGO decay ambient	300 ns	literature	fixed
CsI(Tl) yield, decay	54 000; 730/3340 ns, 64/36%	calibration; literature	fixed
intrinsic resolution	5% FWHM at 511 keV, both crystals, scaling as $(511\ \text{keV}/E)^{1/2}$	our CsI(Tl) data; BGO curve from Gentile et al.; Moszynski et al. to check; the energy dependence is an assumption	BGO to compare with data; exponent to check on the 1275 keV line
emission spectra	tabulated	literature	to digitise
breakdown voltage	$38\ \text{V} + 34\ \text{mV/K}\,(T - 298\ \text{K})$	datasheet; our measurement	measured per run
gain, PDE, crosstalk vs overvoltage	Section 6	datasheet curves	fixed
dark rate	6 MHz at 25 °C, 2.7 V; 10 kHz at 196 K, 6.5 V	datasheet; our measurement	measured per run
afterpulsing	1% at 3 V, $\propto V_{\text{ov}}^2$, 100 ns	literature on the same technology	parameter, not resolvable with the Citiroc
pulse shape	16/24/50 ns into 100 Ω	our measurement with the digitizer	fixed for the digitizer; parameter at the Citiroc input
digitizer scale	3.2 counts per photoelectron per volt of overvoltage at gain 1; area 4.4 times the peak	our measurement	measured per run
digitizer noise	4 counts rms per sample, white	datasheet; our measurement	measured per run; low-frequency and group terms to add
digitizer settings	gain, record, pre-trigger, threshold, majority	our acquisition	setting of the run
gate and baseline window	1 µs (BGO ambient), 4 to 5 µs (CsI(Tl)), 10 µs (cold); 1 µs before the trigger	this document	setting of the analysis
Citiroc shaper	CR-RC^2, $t_p = 87.5$ ns	datasheet	setting of the run
Citiroc noise	0.1 pe rms on the high-gain path	datasheet	measured per run
hold delay	up to 500 ns	our acquisition	setting of the run
ADC	12 bit	assumption	setting of the run
source distance	30 mm, on axis, far face	setup	fixed